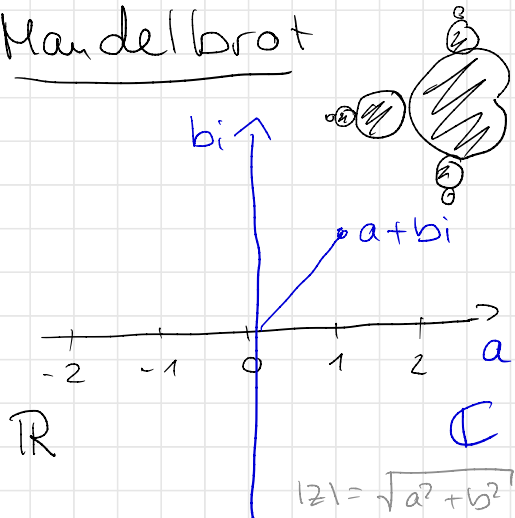


Mandelbrot



→ a set of numbers
 $f_c(z) = z^2 + c$

c is a Complex Number

$$a + bi$$

$$i^2 = -1$$

$$i = \sqrt{-1}$$

$$x^2 + 1 = 0$$

$$z \rightarrow z^2 + c$$

- Choose a c

- z starts at 0, then we iterate

$$c = 1 + 0i = 1$$

$$f_1(0) = 0^2 + 1 = 1$$

$$f_1(1) = 1^2 + 1 = 2$$

$$f_1(2) = 2^2 + 1 = 5$$

$$f_1(5) = 5^2 + 1 = 26$$

$$f_1(26) = 26^2 + 1 = 676$$

$$c = -1 + 0i = -1$$

$$f_{-1}(0) = 0^2 - 1 = -1$$

$$f_{-1}(-1) = (-1)^2 - 1 = 0$$

$$f_{-1}(0) = 0^2 - 1 = -1$$

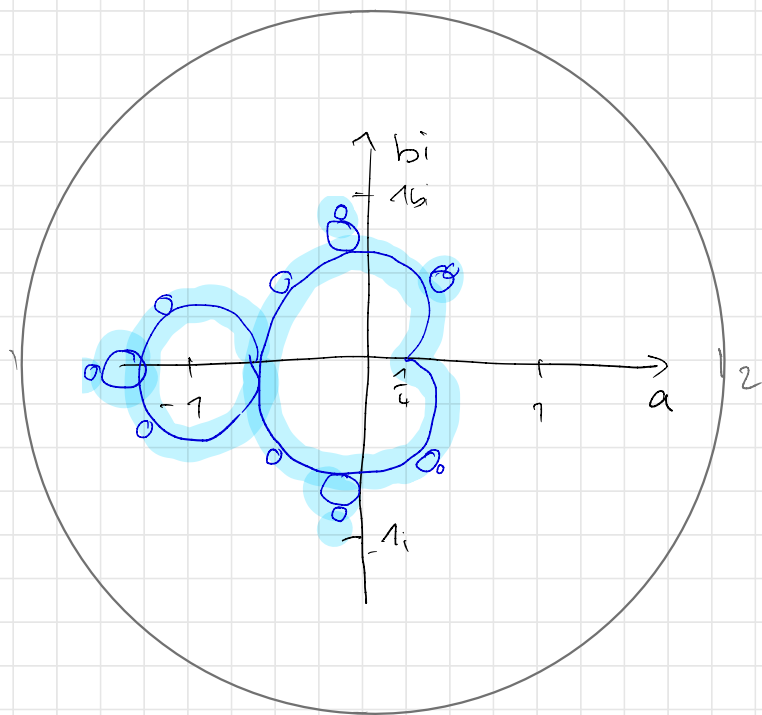
⋮

CASE 1: $|z|$ unbounded
(escapes to infinity)

CASE 2:
 $|z|$ bounded
(≤ 2)

Check for each c in the plane.

The Mandelbrot set M is the set of complex numbers for which the distance $|z|$ (from the origin 0) is bounded.



Coloring: "how quickly f escapes"

Boundaries: - Infinitely detailed at every scale
- Self-similar but not quite repeating
- Formally in computable in places

↳ if you change c a little bit you can't predict what will happen on the boundary, while the insides are completely predictable.

"Square a number add a constant, repeat
- but it encodes 'more visual and mathematical complexity than most objects do.'"

Rendering



Pixel (p_x, p_y) corresponds to which complex number $(a+bi)$?

$$z = a + bi$$

$$c = x + yi$$

$$z^2 = (a + bi)^2$$

$$\Leftrightarrow (a + bi)(a + bi)$$

$$\Leftrightarrow a^2 + abi + abi - b^2$$

$$\Leftrightarrow (a^2 - b^2) + 2abi$$

$$\begin{aligned} (bi)^2 &= b^2 \cdot (-1) \\ &= -b^2 \end{aligned}$$

$$z^2 + c = (a^2 - b^2) + 2abi + x + yi$$

$$\Leftrightarrow \underbrace{(a^2 - b^2 + x)}_{\text{real}} + \underbrace{(2ab + y)i}_{\text{imaginary}}$$

real

imaginary

Pseudo Code

$\left. \begin{array}{l} a = 0 \\ b = 0 \end{array} \right\} \begin{array}{l} x = \dots \\ y = \dots \end{array} \right\} \text{from pixel coordinates}$

for $i = 0$ to max_iterations :

$$a_next = a^2 - b^2 + x$$

$$b_next = 2ab + y$$

if $(a_next^2 + b_next^2) >> 4$) BREAK

$$a = a_next$$

$$b = b_next$$